

# RAG Chatbot — Project Brief

Portfolio project · AI/ML · Owner: Luqman Ahmad

## Profile

|           |                                                         |
|-----------|---------------------------------------------------------|
| Name      | Luqman Ahmad                                            |
| Role      | AI Engineer, Ninjascode (Islamabad, Pakistan)           |
| Education | BS Computer Science, University of Malakand (2022–2026) |
| Location  | Lower Dir, Pakistan                                     |

## Project Goal

Build and ship a working Retrieval-Augmented Generation (RAG) chatbot as a quick, high-quality portfolio piece. Target timeline: 1–2 days. The chatbot should let a user upload PDF/company documents and ask natural-language questions, with answers grounded in the source documents (with citations back to the source text where possible).

## Data Source

PDF documents / company docs. Example use cases: internal policy documents, product manuals, research papers, or a company knowledge base. For the demo, 3–10 representative PDFs (mixed length) are enough to show retrieval quality without overengineering ingestion.

## Tech Stack

- Orchestration: LangChain
- LLM: OpenAI API or Anthropic Claude API (either works with LangChain's model wrappers)
- Embeddings: OpenAI text-embedding-3-small, or a local sentence-transformers model as a free alternative
- Vector store: FAISS or Chroma (both lightweight, no external server needed)
- Document loading: LangChain's PyPDFLoader / UnstructuredPDFLoader
- UI: Streamlit or Gradio for a quick, demoable front end

## Architecture (Pipeline Stages)

1. Load PDFs and split into overlapping chunks (e.g. 500–1000 tokens, 10–20% overlap).
2. Embed chunks and store vectors in FAISS/Chroma with metadata (source file, page number).
3. On a user query: embed the query, retrieve top-k relevant chunks.
4. Construct a grounded prompt (retrieved chunks + question) and call the LLM via LangChain.

- 5. Return the answer with source citations; display in a Streamlit/Gradio chat UI.

## **Deliverables for Portfolio**

- Working demo (local script or deployed Streamlit/Gradio app)
- Clean GitHub repo with README (setup steps, architecture diagram, example queries)
- Short write-up or demo video/GIF showing a question being answered with citations

Notes: This brief is meant as project context for Claude — upload it to the project's Context section so future chats in this project already know the goal, data source, and stack without needing to be re-explained.